

Biological Image Analysis Using Fiji

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In the last couple of issues we dealt with biomedical signal-processing software, namely, BioSig and FieldTrip. Fiji Is Just ImageJ software, abbreviated as Fiji, is a smart open source platform for the analysis of biomedical data. As the name indicates, it is a distribution of ImageJ package with many useful plugins. The latest version of this software

licensed under GNU public licence version 2 is bundled in the DVD accompanying this month's *EFY Plus*.

Why Fiji

Recent advances in computer vision and technology have helped immensely in the analysis of biological images. Computerised analysis of images makes it possible to overcome the limitations of human observations, providing relevant samples or high-resolution information of smaller samples. Image data is registered from various overlapping parts, segmented to cap-

ture biologically-relevant features, tracked across space and time, and compared with other specimen.

There are a handful of software in the market that deal with biological images. These include commercial ones like Amira and Velocity and open source ones including ImageJ and BioImageXD. While commercial platforms provide an easy interface, inner details of image-processing algorithms are hidden from the user, which is not recommended when it comes to research in bio-image informatics.

If we take the case of an open source software like ImageJ, it is made by and for biologists. Therefore the tool does not follow the principles of modern software engineering, making it less attractive for computer scientists.

Project Fiji is built on the architecture of ImageJ, with an aim to help researchers to develop innovative, cutting-edge solutions for biological-image analysis. Powerful libraries in Fiji ensure that image-analysis tools are frequently updated with the latest algorithms.

Attracting researchers from multiple disciplines

The software proves to be a very useful tool for researchers from various disciplines. If you are a researcher in biology, the point-and-click interface provided by Fiji allows you to interact with multi-dimensional data. This interface is quite similar to the one provided by ImageJ. A bio-informatician with sound programming skills can make use of the available scripting languages for constructing image-processing pipelines. This is possible with the help of Script Editor plugin.

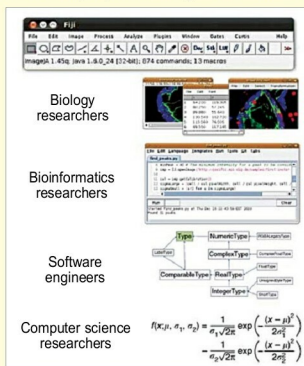


Fig. 1: Fiji is a truly interdisciplinary software connecting engineers and researchers from various disciplines

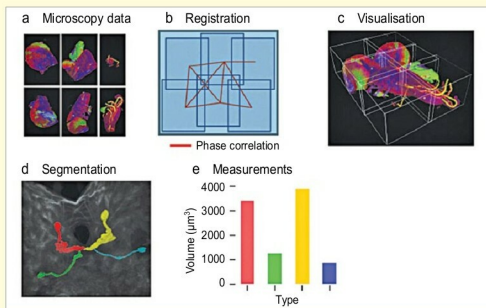


Fig. 2: Some of the main techniques used in the software (a) Stitching plugins helps in gathering different views of the microscopic data, (b) Image registration, (c) Visualisation, (d) Image segmentation and (e) Measurements (Image courtesy: Article titled 'Fiji: An open source platform for biological-image analysis' by Johannes Schindelin et al)